SECURITY INCIDENT MANAGEMENT & AUTOMATED RESPONSE WORKFLOW

Project Duration: September 2026
Project Category: SOC Operations | ServiceNow Development
Project Type: Self-Paced Personal Lab
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1. PROJECT OVERVIEW

Designed a practical Security Operations Center (SOC) incident management workflow integrating Windows security event monitoring, Splunk SIEM detection and a custom ServiceNow incident tracking application.

The project focuses on identifying suspicious authentication activities, investigating security alerts, classifying incidents and managing their lifecycle through structured workflows.

2. KEY TECHNICAL SKILLS

- **SIEM Operations:** Splunk Enterprise, SPL Queries, Security Event Monitoring.
- **Windows Security:** Event ID 4624, Event ID 4625, Windows Event Viewer.
- **ServiceNow Development:** Custom Tables, Forms, Incident Records, Application Development.
- **Workflow Automation:** Flow Designer, Assignment Rules, Priority Management.
- **SOC Operations:** Alert Triage, Log Analysis, Incident Investigation, Incident Response.
- **Security Frameworks:** MITRE ATT&CK, Incident Management Lifecycle.
- **Project Delivery:** Agile Methodology, Technical Documentation, GitHub.

3. PROJECT OBJECTIVES

- Detect suspicious authentication activities through Windows security logs.
- Identify repeated failed login attempts using Splunk.
- Investigate security alerts using relevant event information.
- Develop a custom ServiceNow security incident management application.
- Configure incident classification, assignment and prioritisation.
- Implement automated incident lifecycle workflows.

4. TECHNOLOGY STACK

Technology	Implementation
Windows 10/11	Authentication event generation
Splunk Enterprise	SIEM monitoring and detection
ServiceNow PDI	Custom incident management application

Flow Designer	Workflow automation
SPL	Security detection queries
MITRE ATT&CK	Brute-force technique reference
GitHub	Project documentation

5. IMPLEMENTATION & TECHNICAL APPROACH

Phase 1: Windows Security Event Monitoring

Configure Windows security auditing and monitor authentication events.

- Event ID 4624 – Successful Logon.
- Event ID 4625 – Failed Logon.

Phase 2: SIEM Detection Using Splunk

Collect Windows security events and develop SPL queries to identify repeated authentication failures.

Sample Detection Query:

index=* EventCode=4625

| timechart span=5m count AS failed_logins

Key Skills: Log Collection | SPL | Event Correlation | SIEM Monitoring

Phase 3: Security Alert Investigation

Investigate suspicious login patterns by analysing:

- Username and account information.
- Event timestamps.
- Source IP address, when available.
- Frequency of failed authentication attempts.
- Related successful login events.

Map relevant brute-force activity to **MITRE ATT&CK T1110 – Brute Force**, where supported by the evidence.

Key Skills: Alert Triage | Threat Analysis | IOC Investigation | MITRE ATT&CK

Phase 4: ServiceNow Incident Application Development

Develop a custom application to record and manage security incidents.

Incident Form Fields:

Field	Purpose
Incident Number	Unique reference
Short Description	Security alert summary
Alert Source	Splunk / Windows
Affected User	Associated account
Severity	Risk classification
Priority	Incident urgency
Assigned To	Responsible analyst
State	New / In Progress / Resolved
Resolution Notes	Investigation outcome

Key Skills: ServiceNow Tables | Forms | Application Development | Incident Management

Phase 5: Workflow Automation

Use Flow Designer to implement:

- Automatic incident assignment.
- Priority updates based on severity.
- Incident status management.
- Resolution and closure tracking.

Key Skills: Flow Designer | Workflow Configuration | Process Automation

Phase 6: Testing and Validation

Generate test failed login events and validate the complete workflow.

Verify:

- Windows event generation.
- Splunk event ingestion.
- Detection query results.
- Incident record creation.
- Assignment and priority updates.
- Incident resolution.

6. EXPECTED PROJECT OUTPUT

- Functional Windows security event monitoring.
- Splunk failed-login detection.
- Custom ServiceNow incident tracking application.
- Automated incident assignment workflow.
- Incident investigation documentation.
- Security monitoring dashboard.

7. PROJECT DELIVERABLES

1. Splunk SPL detection queries.
2. Windows Event Viewer screenshots.
3. Splunk security dashboard.
4. ServiceNow incident application screenshots.
5. Flow Designer workflow documentation.
6. Sample incident investigation report.

8. BUSINESS & SECURITY VALUE

Demonstrates how SIEM monitoring and structured incident management can improve security visibility, investigation consistency, incident accountability and response coordination.

9. LEARNING OUTCOMES

Practical understanding of SOC operations, SIEM detection, Windows authentication events, ServiceNow application development, workflow automation and incident response processes.

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